ISMAEL ESPINOZA JARAMILLO

Robot Learning Researcher | Sim-to-Real Policy Deployment | Vision-Based Dexterous Manipulation | Reinforcement Learning

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Google Scholar: scholar.google.com/citations?user=3S9BoVAAAAAJ

Summary

Robot learning researcher (Ph.D. expected Feb 2027, Kyung Hee University) who trains manipulation policies in simulation and deploys them on real robots. Built an end-to-end Real2Sim2Real system (teleoperated data collection, Isaac Lab digital twin, transformer reinforcement learning, RGB visuomotor policy distillation, ROS 2 deployment) that runs 44-DoF two-arm, two-hand policies zero-shot on a table-mounted and an upper-humanoid robot. Hands-on with imitation learning and VLA models (ACT, SmolVLA). 4 journal articles (incl. IEEE RA-L), 7 international conference papers (ICRA, HRI, EMBC, UR), 1 granted patent, 150+ citations.

Experience

Graduate Researcher (Ph.D.), Robot Learning Mar 2022 – Present

Dept. of Electronics and Information Convergence Engineering, Kyung Hee University, Yongin, South Korea | Advisor: Prof. Tae-Seong Kim

- Designed and built a Real2Sim2Real system that trains 44-DoF bimanual dexterous policies (UR3e and UR3 arms, two 16-DoF Allegro Hands) in an Isaac Lab digital twin and deploys them zero-shot on two robot platforms (table-mounted and upper-humanoid), with no training on real-robot data.
- Built the digital twin from phone-scanned object meshes and actuator gains calibrated against recorded hardware motion; trained under automatic domain randomization with 500 parallel GPU environments per run.
- Built the teleoperation data pipeline (Manus gloves, VIVE trackers, hand retargeting) and a trajectory-stitching augmentation that turns 30 human demonstrations into 300 physically validated demonstrations per task.
- Co-developed TC-TRL, a temporal-context multi-agent transformer policy pretrained by behavior cloning and fine-tuned with PPO (IEEE RA-L 2024, equal-contribution first author; granted Korean patent): 89.9% mean success on five long-horizon tasks in simulation vs 59.0% for the best transformer baseline and 26.6% for the best PPO baseline.
- Deployed the policies through a ROS 2 bridge with FoundationPose 6D object tracking on Intel RealSense cameras: four long-horizon real-robot tasks, one with a hand-to-hand handover, succeeded in 18 to 30 of 35 trials per task, losing 5 to 11 points against simulation where behavior cloning alone lost up to 23.
- Co-developed TMAD, online teacher-student distillation into an RGB-only visuomotor policy (two cameras, 30 Hz; no pose estimator, object model, or camera calibration): 72 to 84% real-robot success on four tasks, 44 to 60 points above a single-agent DextrAH-RGB-style student on three of them.
- Earlier: two first-author papers (Sensors) on deep-learning human activity recognition and forecasting from wearable exoskeleton IMU signals, 120+ citations combined.

Research Assistant, Mobile Robotics May 2021 – Feb 2022

Universidad de las Fuerzas Armadas (ESPE), Quito, Ecuador

- Built the ROS software stack (Python, C++) for a social-navigation and telepresence mobile robot, running deep-learning perception and navigation in real time on board, plus BLE indoor localization, low-level electronics, and PID wheel control.

Project Development Assistant, Industrial Automation Nov 2020 – May 2021

Robotron S.A., Quito, Ecuador

- Built a LiDAR and QR-marker forklift navigation and pallet-verification system for a sub-zero food warehouse.
- Wrote C++ and Python bridges between Siemens PLCs (TIA Portal, OPC), the navigation software, and a Microsoft Azure database; designed the control panels and installed panels, sensors, and wiring on site.

Earlier: Electrical Design Intern, SEIUS S.A. (2020); Teaching Assistant, Electronics II, ESPE (2018).

Selected Projects (open source)

Dexterous Pick-and-Place with Grasp Synthesis – github.com/ismael-esp/dexterous-pick-and-place 2026

- Force-closure grasp synthesis (BODex on cuRobo) and FoundationPose perception driving both Isaac Lab and a real UR3 with an Allegro Hand: 94 of 100 executed real-robot trials succeeded across 10 objects; 180 of 200 episodes in simulation across both arms.

FoundationPose ROS 2 Tracker – github.com/ismael-esp/foundationpose-ros2-tracker 2026

- Zero-shot 6D object tracking in ROS 2 from a mesh and an object name (YOLO-World, SAM 2, FoundationPose), publishing world-frame poses for manipulation; ported the CUDA extensions to Blackwell GPUs (PyTorch 2.11, CUDA 12.8).

FoldingACT, IROS 2026 Robotic Origami Challenge – github.com/ismael-esp/robotic-origami-challenge 2026

- Action-chunking transformer (ACT-style) imitation policy fusing four cameras (DINOv3) and ten fingertip tactile sensors (Sparsh) for bimanual paper folding with two 22-joint hands (65-D actions), trained on a 398 GB LeRobot dataset; built the data pipeline, an Isaac Sim mirror, and the inference service the organizers ran on their robot.

Patents

Korean Patent 10-2943203 (granted Mar 2026). T.-S. Kim, **I. Espinoza Jaramillo**, J.-H. Oh. System and method for long-horizon task execution of a dual-arm robot using temporal-context transformer network-based reinforcement learning. Two further Korean patent applications pending (2025, 2026).

Selected Publications

J.-H. Oh*, **I. Espinoza***, D. Jung, T.-S. Kim. Bimanual long-horizon manipulation via temporal-context transformer RL. *IEEE Robotics and Automation Letters*, 9(12), 2024. Presented at IROS 2025. (*equal contribution)

I. Espinoza, Y.-H. Choi, K. Lee, J.-H. Oh, K. J. Pahk, T.-S. Kim. A Real2Sim2Real transformer policy system for bimanual R2R2H tasks. *ACM/IEEE HRI 2026 Companion*.

Y.-H. Choi, **I. Espinoza**, K. Lee, S. Park, J. Y. Kim, J.-H. Oh, W. H. Lee, T.-S. Kim. A semi-humanoid Real2Sim2Real system for bimanual long-horizon manipulation. *Int. Conf. on Ubiquitous Robots (UR)*, 2026.

S. Park, **I. Espinoza**, G. Kim, J. Y. Kim, K. Lee, T.-S. Kim. TMAD: Temporal multi-agent distillation for bimanual long-horizon manipulation. Under review, 2026.

J.-H. Oh, D. Jung, **I. Espinoza**, et al. Multi-modal affordance planner with temporal-context action policy for long-horizon bimanual robot manipulation. *IEEE ICRA*, 2026.

J.-H. Oh*, **I. Espinoza***, et al. Bimanual long-horizon lifecare robotics with temporal context LLM planner and transformer RL. *IEEE EMBC*, 2025. (*equal contribution)

I. Espinoza Jaramillo, J. G. Jeong, et al. Real-time human activity recognition with IMU and encoder sensors in wearable exoskeleton robot via deep learning networks. *Sensors*, 22(24), 2022.

Also 2 journal articles, 3 international conference papers, 1 symposium poster, and 6 domestic conference papers; full list on Google Scholar.

Skills

Robot learning: reinforcement learning (PPO, multi-agent RL), imitation learning and behavior cloning, VLA and action-chunking policies (ACT, SmolVLA), transformer policies, teacher-student policy distillation, sim-to-real transfer, domain randomization, system identification, teleoperation data collection
Perception: 6D object pose estimation and tracking (FoundationPose), open-vocabulary detection and segmentation (YOLO-World, SAM 2), RGB visuomotor policies, multi-camera RGB-D (Intel RealSense)
Simulation: NVIDIA Isaac Lab, Isaac Sim, Isaac Gym, MuJoCo, digital twins, massively parallel GPU training
Software: Python, C++, PyTorch, CUDA, ROS 2, Docker, Linux, Git, LeRobot, cuRobo
Hardware: UR3 and UR3e arms, Allegro Hand V4, Manus gloves, VIVE trackers, Siemens PLCs, LiDAR

Education

Ph.D. (integrated M.S.-Ph.D.), Electronics and Information Convergence Engineering (major: Biomedical Engineering)
Mar 2022 – Feb 2027 (expected)
Kyung Hee University, South Korea | GPA 4.27/4.30

Dissertation: A Real2Sim2Real System for Bimanual Long-Horizon Dexterous Manipulation via Temporal-Context Transformer Reinforcement Learning

B.Sc., Electronics, Automation and Control Engineering Sep 2015 – Feb 2021
Universidad de las Fuerzas Armadas (ESPE), Ecuador | GPA 17.15/20

Honors

President's Scholarship, Kyung Hee University, all 6 semesters (2022–2024) | BK21 FOUR Fellowship (2025) | BK21 Plus Scholarship (2023–2024) | Journal Publication Scholarship, Kyung Hee University (2023)

Languages

Spanish (native) | English (fluent) | Korean (basic)